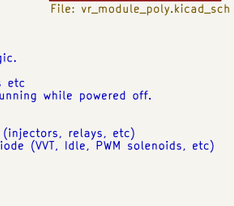


16p header placement markers

J6 P1x16 J9 P1x16 J7 P1x16 J10 P1x16 J8 P1x16 J11 P1x16

5p header placement markers

J12 P1x5 J13 P1x5



PIN FUNCTIONS

12V_KEY – Connect to ignition switch. Powers internal ECU logic.
12V_MR – Power input from the ECU-controlled main relay.
12V_MR_FUSE – Internally-fused 12V power output for sensors etc
PERM_LIVE – 12v permanent battery connection, keeps clock running while powered off.
5V_SENSOR – 5V supply output for ECU sensors

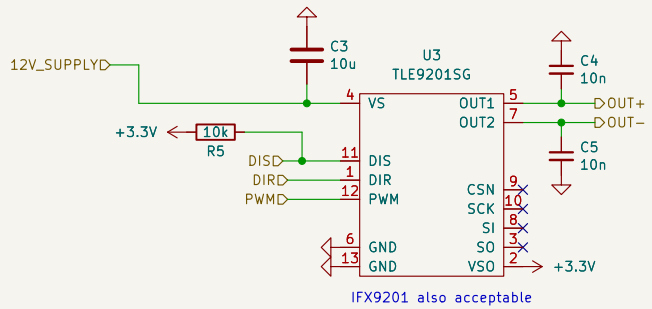
LS1-8, LS13-16 – Medium power low-side outputs, no diode (injectors, relays, etc)
LS9-12, A-D – High power low-side outputs with freewheel diode (VVT, Idle, PWM solenoids, etc)
HS1-4 – Medium power high side outputs

happycactusgarage.com
FutureProofPerformance.com
github.com/FOME-Tech

Sheet: /
File: Polygonus-Universal-Base.kicad_sch

Title: FOME Universal 112 pin ECU base

Size: A3	Date: 2026-03-03	**Rev: v1.2**
KiCad E.D.A. 10.0.7		Id: 1/10



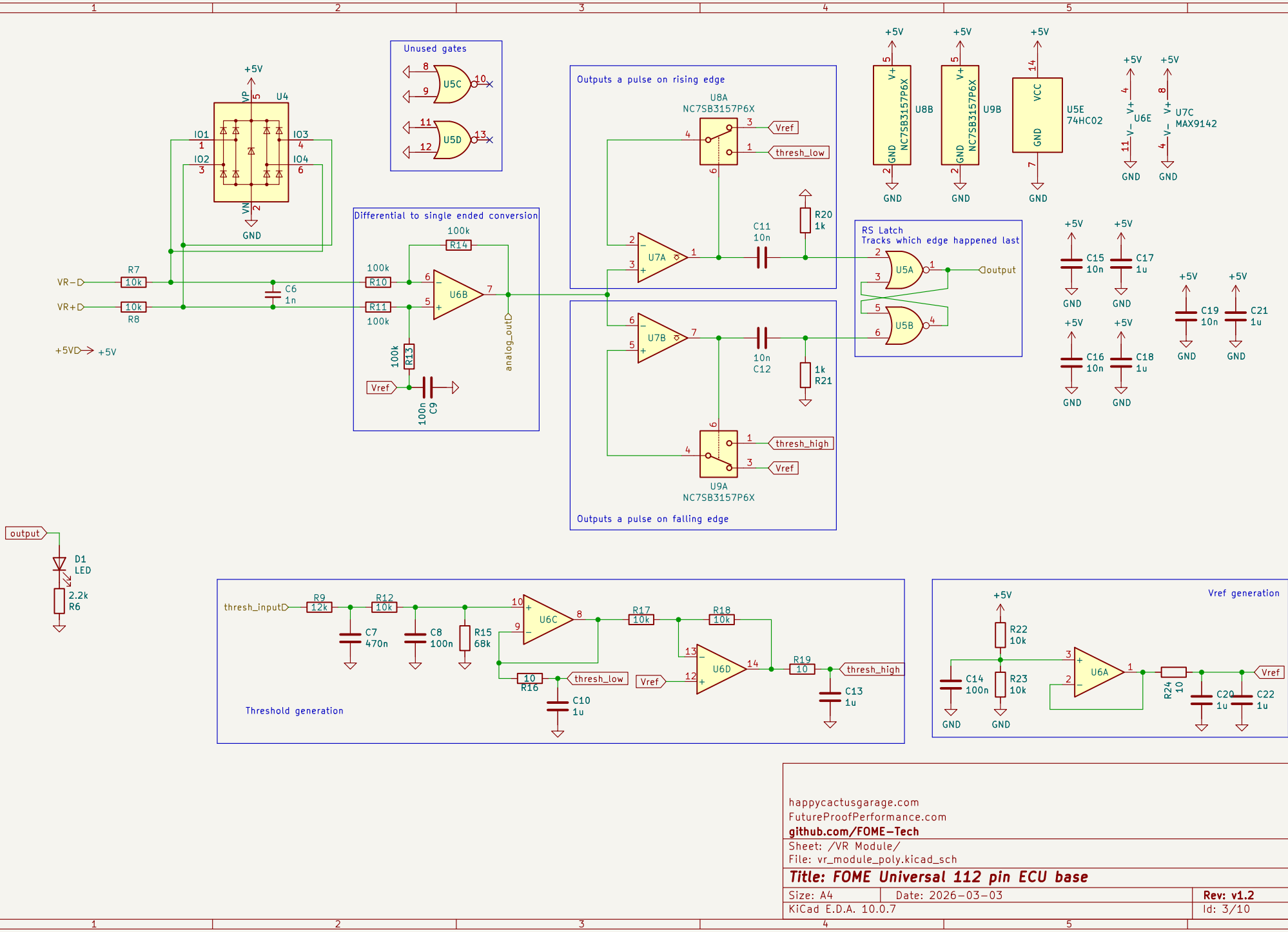
IFX9201 also acceptable

happycactusgarage.com
FutureProofPerformance.com
github.com/FOME-Tech

Sheet: /etb-2/
File: etb-9201.kicad_sch

Title: FOME Universal 112 pin ECU base

Size: A4	Date: 2026-03-03	**Rev: v1.2**
KiCad E.D.A. 10.0.7		Id: 2/10

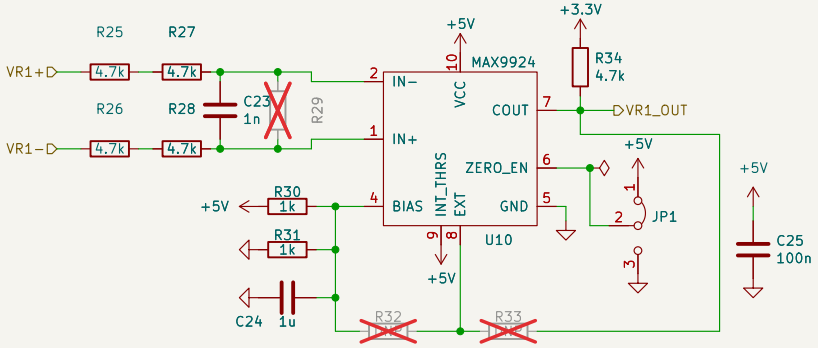


happycactusgarage.com	
FutureProofPerformance.com	
github.com/FOME-Tech	
Sheet: /VR Module/	
File: vr_module_poly.kicad_sch	
Title: FOME Universal 112 pin ECU base	
Size: A4	Date: 2026-03-03
KiCad E.D.A. 10.0.7	Id: 3/10

Rev: v1.2

Yes, we know that the polarity is flipped going in to the VR interface chips.
While originally on accident, it's no big deal since they invert anyway:
When VR+ transitions to a higher voltage than VR−, VR_OUT will output a rising edge.

See issue https://github.com/mck1117/proteus/issues/57 for more detail



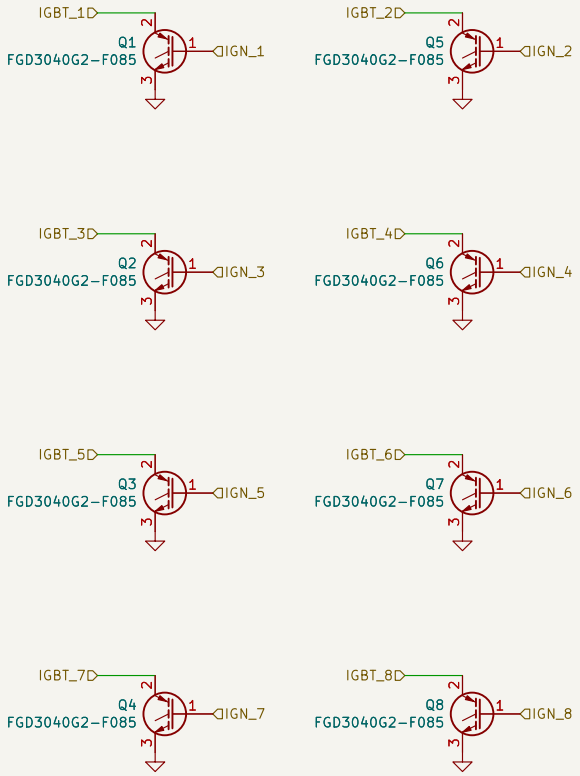
Optional setting in Mode C for no adaptive threshold timeout − See MAX9924 Datasheet
Components as specified put chip in mode A1. Flip jumper and install R1219+R2020 for mode C.

happycactusgarage.com
FutureProofPerformance.com
github.com/FOME-Tech

Sheet: /VR Conditioner/
File: VR Conditioner.kicad_sch

Title: FOME Universal 112 pin ECU base

Size: A4	Date: 2026-03-03	**Rev: v1.2**
KiCad E.D.A. 10.0.7		Id: 4/10

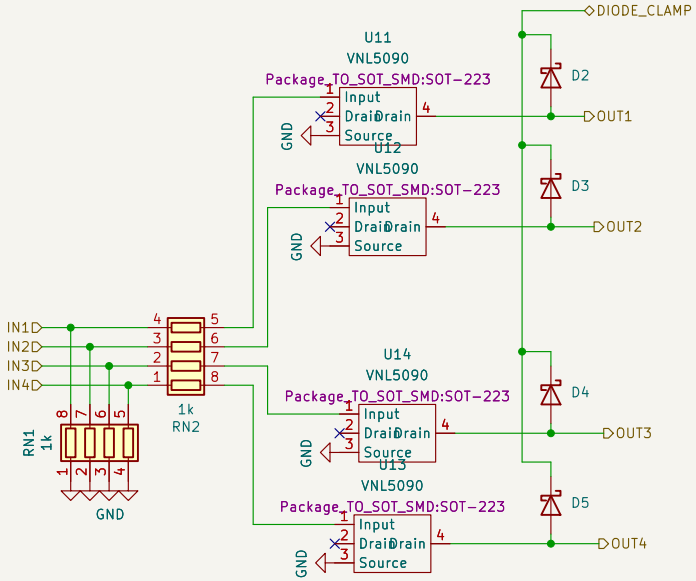


happycactusgarage.com
FutureProofPerformance.com
github.com/FOME-Tech

Sheet: /IGBT outputs/
File: IGBT outputs.kicad_sch

Title: FOME Universal 112 pin ECU base

Size: A4	Date: 2026-03-03	**Rev: v1.2**
KiCad E.D.A. 10.0.7		Id: 5/10

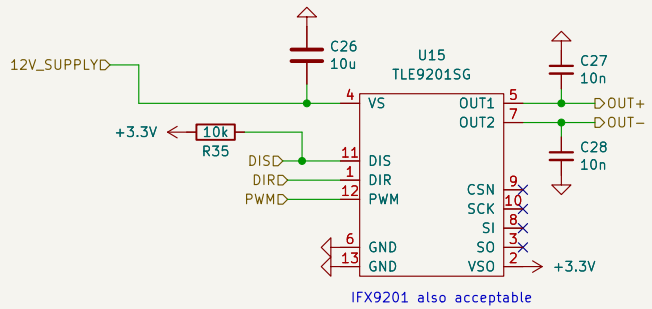


happycactusgarage.com
FutureProofPerformance.com
github.com/FOME-Tech

Sheet: /Lowside Adder/
File: Lowside Adder.kicad_sch

Title: FOME Universal 112 pin ECU base

Size: A4	Date: 2026-03-03	**Rev: v1.2**
KiCad E.D.A. 10.0.7		Id: 6/10

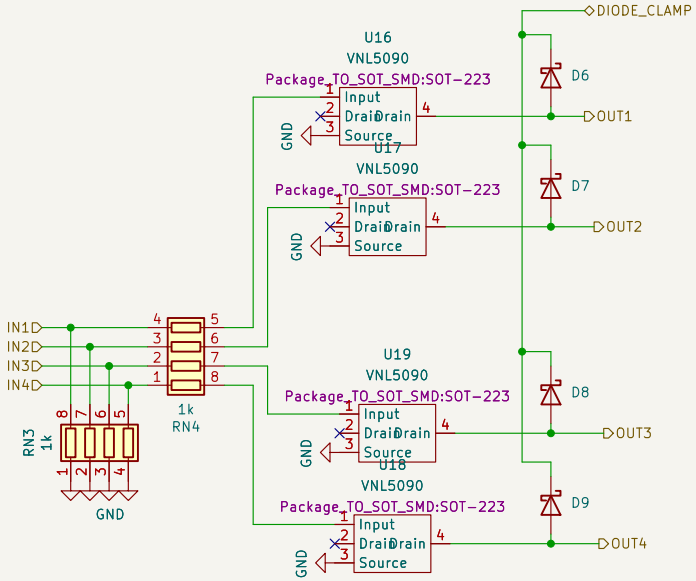


happycactusgarage.com
FutureProofPerformance.com
github.com/FOME-Tech

Sheet: /etb-1/
File: etb-9201.kicad_sch

Title: FOME Universal 112 pin ECU base

Size: A4	Date: 2026-03-03	**Rev: v1.2**
KiCad E.D.A. 10.0.7		Id: 7/10

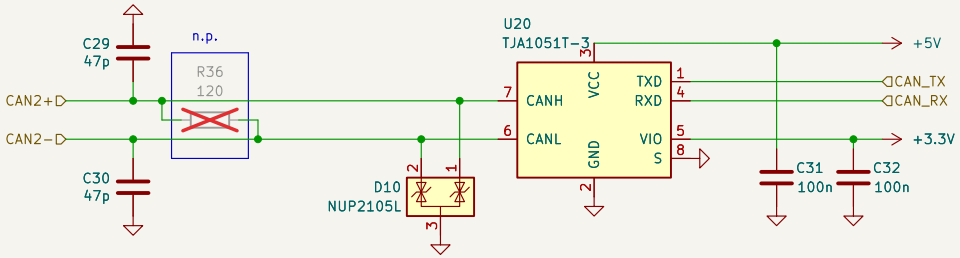


happycactusgarage.com
FutureProofPerformance.com
github.com/FOME-Tech

Sheet: /Lowside Adder1/
File: Lowside Adder.kicad_sch

Title: FOME Universal 112 pin ECU base

Size: A4	Date: 2026-03-03	**Rev: v1.2**
KiCad E.D.A. 10.0.7		Id: 8/10

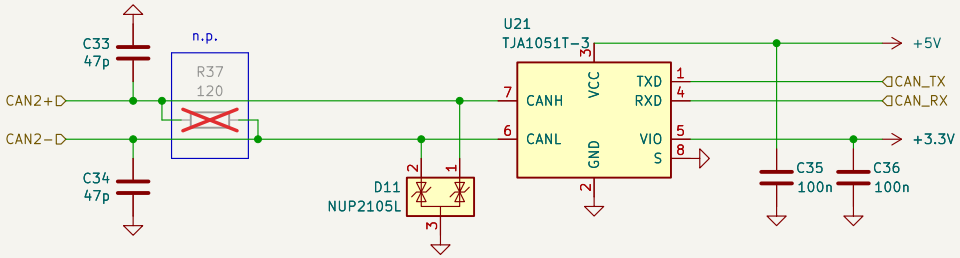


happycactusgarage.com
FutureProofPerformance.com
github.com/FOME-Tech

Sheet: /WBO-CAN-TRANS/
File: wbo-can-trans.kicad_sch

Title: FOME Universal 112 pin ECU base

Size: A4	Date: 2026-03-03	**Rev: v1.2**
KiCad E.D.A. 10.0.7		Id: 9/10



happycactusgarage.com
FutureProofPerformance.com
github.com/FOME-Tech

Sheet: /WBO-CAN-TRANS1/
File: wbo-can-trans.kicad_sch

Title: FOME Universal 112 pin ECU base

Size: A4	Date: 2026-03-03	**Rev: v1.2**
KiCad E.D.A. 10.0.7		Id: 10/10